

Comparing Categorical Frequencies in a Two-Way Table

Reading a two-way table, row and column relative frequencies, and comparing them to judge association.

All ten problems use the survey table below. Two hundred eighth graders were asked two yes-or-no questions: *Do you play a school sport?* and *Do you get 8 or more hours of sleep on a school night?* Every number in the table is a count of students.

- A **row relative frequency** divides a cell by *its own row* total. A **column relative frequency** divides the same cell by *its own column* total. They answer different questions, so always say which total you used.
- Give every percent to the nearest whole percent.

	8 or more hours	Under 8 hours	Row total
Plays a school sport	30	20	50
Does not play a sport	45	105	150
Column total	75	125	200

1. How many of the 200 students both play a school sport and get 8 or more hours of sleep? Name the row and the column you used to find the cell.

Answer: _____

2. Add the four inner cells of the table — 30, 20, 45 and 105 — to check the grand total. How many students were surveyed altogether?

Answer: _____

3. Of the 50 students who play a school sport, 30 get 8 or more hours of sleep. What percent of the sport players get 8 or more hours? Write the row relative frequency as a percent.

Answer: _____

4. Find how many more of the students who do not play a sport get under 8 hours of sleep (105 students) than the students who do play a sport and get under 8 hours (20 students).

Answer: _____

5. Of the 150 students who do not play a school sport, 45 get 8 or more hours of sleep. What percent of the non-players get 8 or more hours? Write the row relative frequency as a percent.

Answer: _____

6. Now switch to a column. Of the 75 students who get 8 or more hours of sleep, 30 play a school sport. What percent of the well-rested students play a sport? Write the column relative frequency as a percent, and state clearly which total you divided by.

Answer: _____

7. Stay in the columns. Of the 125 students who get under 8 hours of sleep, 20 play a school sport. What percent of the short-sleep students play a sport? Write the column relative frequency as a percent, and state which total you divided by.

Answer: _____

8. Problem 3 gave the percent of sport players who get 8 or more hours of sleep (60 percent) and problem 5 gave the same percent for non-players (30 percent). Subtract to find the gap between the two row relative frequencies, in percentage points.

Answer: _____

9. Five other eighth-grade classes ran the same survey and built their own two-way tables. The percent of sport players getting 8 or more hours of sleep came out as 55, 70, 45, 60 and 60. Find the median of these five percents.

Answer: _____

10. Use your answers from problems 3 and 5. Is there an association between playing a school sport and getting 8 or more hours of sleep? Say yes or no, and then explain in complete sentences why comparing the two row percentages is better evidence than comparing the raw counts 30 and 45.